Statistics for Computer Scientists

Lecture 9: Associations between categorical variables

14 November 2025

Universiteit Leiden
The Netherlands

Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Tba	
12 Dec	Tba	

This lecture

- Chi-squared test of independence
- Dependent vs independent variables
- Patterns of association
- Measures of association

Chi-squared test of independence

Universiteit Leiden
The Netherlands

Two categorical variables

- Use when both response and explanatory variables are **categorical**
  - I.e., we are still considering bivariate settings
- Two groups and two outcome options: z-test for two samples/groups

- $\hat{\pi}_1 = \frac{272}{321} = 0.85$
- $\hat{\pi}_2 = \frac{185}{393} = 0.47$

- $H_0: \pi_2 = \pi_1$
- $H_a: \pi_2 \neq \pi_1$

		Happiness		
		Very	Not too	*Total*
Income	Above avg	272	49	321
	Below avg	185	208	393
	Total	457	257	714

Two categorical variables

- Use when both response and explanatory variables are **categorical**
- More than 2 groups and/or outcome options: **chi-squared test**

- There is an **association** if the conditional population distribution for the response variable differs among categories of the explanatory variable

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	272	294	49	615
	Average	454	835	131	1420
	Below avg	185	527	208	920
	Total	911	1656	388	2955

Association in the population?

Dependence vs independence

- Previous lecture: dependent vs independent samples (groups)
- Dependence for variables:

- **Statistical independence**: population conditional distributions on one variable are the same for all categories of the other variables (no association)
- **Statistical dependence**: conditional distributions are not all identical (association)

Example of independence

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	32%	55%	13%	615
	Average	32%	55%	13%	1420
	Below avg	32%	55%	13%	920
	Total	911	1656	388	2955

Independence or dependence?

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	44%	48%	8%	615
	Average	32%	59%	9%	1420
	Below avg	20%	57%	23%	920
	Total	911	1656	388	2955

Chi-squared test of independence

1. **Assumptions**

   Two categorical variables

   Random sampling

   Expected count in each cell is at least 5

2. **Hypotheses**

   H_0: variables are statistically independent

   H_a: variables are statistically dependent

Chi-squared test of independence

3. **Test statistic**

Intuition: summarize the differences between *observed* cell counts and *expected* cell counts (if H_0 were true)

Notation: f_o = observed frequency

f_e = expected frequency

r = number of rows, c = number of columns

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e},$$

with sum taken over all cells in the table

Expected frequencies…

- …have identical conditional distributions, that are also identical to the marginal distribution of the response variable

- …have the same marginal distributions (row and column totals) as the observed frequencies

- Computed by $f_e = (row\ total) * (column\ total)/n$

Expected frequencies

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	f_e?			615
	Average				1420
	Below avg				920
	Total	911 (30.8%)	1656 (56.0%)	388 (13.1%)	2955

First cell: $f_e = (row\ total) * \frac{(column\ total)}{n} = 615 * \frac{911}{2955} = 189.6$

Expected frequencies

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	190 (30.8%)	345 (56.0%)	81 (13.1%)	615
	Average	438 (30.8%)	796 (56.0%)	187 (13.1%)	1420
	Below avg	284 (30.8%)	516 (56.0%)	121 (13.1%)	920
	Total	911 (30.8%)	1656 (56.0%)	388 (13.1%)	2955

Observed vs (expected) frequencies

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	272 (190)	294 (345)	49 (81)	615
	Average	454 (438)	835 (796)	131 (187)	1420
	Below avg	185 (284)	527 (516)	208 (121)	920
	Total	911	1656	388	2955

- Each cell contains f_o and f_e (in parentheses)
- Are differences significant?

Observed vs (expected) frequencies

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	272 (190)	294 (345)	49 (81)	615
	Average	454 (438)	835 (796)	131 (187)	1420
	Below avg	185 (284)	527 (516)	208 (121)	920
	Total	911	1656	388	2955

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e} = \frac{(272 - 189.6)^2}{189.6} + \frac{(454 - 437.8)^2}{437.8} + \cdots$$

$$= 172.3$$

Chi-squared test of independence

4. **P-value**

How are χ^2 values distributed? Standard normal distribution for z-values, t distribution for t-values…

Different distribution! χ^2 follows the chi-squared distribution

Chi-squared distribution

- Positive values only
- Skewed right
  - More bell-shaped as *df* increases
- $df = (r-1)(c-1)$
- Mean: $\mu = df$
- Standard deviation: $\sigma = \sqrt{2df}$
- **Larger χ^2 means lower p-value**

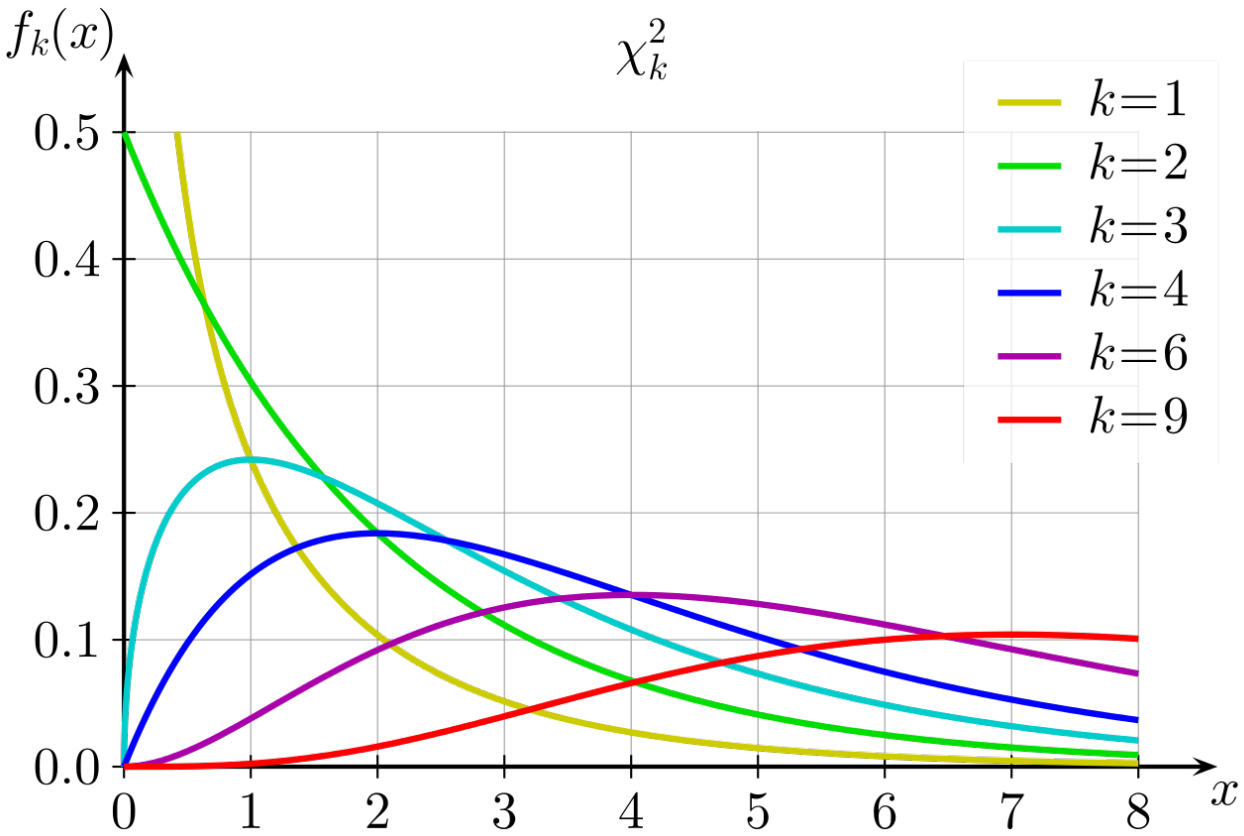


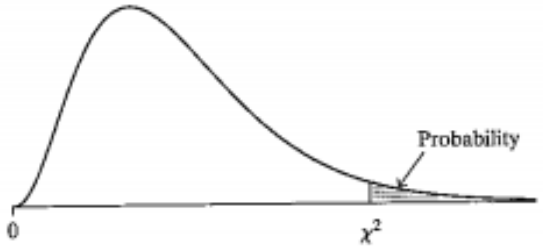
Table C

- Like Table B: *df* on rows, α on columns, cells contain critical values

- Happiness example: $\chi^2 = 172.3$
- $df = (r-1)(c-1) = 2 * 2 = 4$

- Calculated χ^2 is larger than all critical values
- P < 0.001

TABLE C: Chi-Squared Distribution Values for Various Right-Tail Probabilities



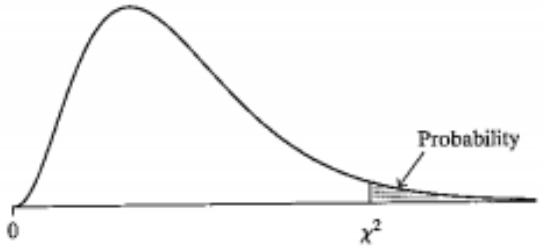
	Right-Tail Probability						
df	0.250	0.100	0.050	0.025	0.010	0.005	0.001
1	1.32	2.71	3.84	5.02	6.63	7.88	10.83
2	2.77	4.61	5.99	7.38	9.21	10.60	13.82
3	4.11	6.25	7.81	9.35	11.34	12.84	16.27
4	5.39	7.78	9.49	11.14	13.28	14.86	18.47
5	6.63	9.24	11.07	12.83	15.09	16.75	20.52
6	7.84	10.64	12.59	14.45	16.81	18.55	22.46
7	9.04	12.02	14.07	16.01	18.48	20.28	24.32
8	10.22	13.36	15.51	17.53	20.09	21.96	26.12
9	11.39	14.68	16.92	19.02	21.67	23.59	27.88
10	12.55	15.99	18.31	20.48	23.21	25.19	29.59
11	13.70	17.28	19.68	21.92	24.72	26.76	31.26
12	14.85	18.55	21.03	23.34	26.22	28.30	32.91
13	15.98	19.81	22.36	24.74	27.69	29.82	34.53
14	17.12	21.06	23.68	26.12	29.14	31.32	36.12
15	18.25	22.31	25.00	27.49	30.58	32.80	37.70
16	19.37	23.54	26.30	28.85	32.00	34.27	39.25
17	20.49	24.77	27.59	30.19	33.41	35.72	40.79
18	21.60	25.99	28.87	31.53	34.81	37.16	42.31
19	22.72	27.20	30.14	32.85	36.19	38.58	43.82
20	23.83	28.41	31.41	34.17	37.57	40.00	45.32
25	29.34	34.38	37.65	40.65	44.31	46.93	52.62
30	34.80	40.26	43.77	46.98	50.89	53.67	59.70
40	45.62	51.80	55.76	59.34	63.69	66.77	73.40
50	56.33	63.17	67.50	71.42	76.15	79.49	86.66
60	66.98	74.40	79.08	83.30	88.38	91.95	99.61
70	77.58	85.53	90.53	95.02	100.4	104.2	112.3
80	88.13	96.58	101.8	106.6	112.3	116.3	124.8
90	98.65	107.6	113.1	118.1	124.1	128.3	137.2
100	109.1	118.5	124.3	129.6	135.8	140.2	149.5

Classified as Internal | Intern

Table C

- P-value is equal to right-tail probability
- No one-sided test option for chi-squared (we square the differences!)

TABLE C: Chi-Squared Distribution Values for Various Right-Tail Probabilities



	Right-Tail Probability						
df	0.250	0.100	0.050	0.025	0.010	0.005	0.001
1	1.32	2.71	3.84	5.02	6.63	7.88	10.83
2	2.77	4.61	5.99	7.38	9.21	10.60	13.82
3	4.11	6.25	7.81	9.35	11.34	12.84	16.27
4	5.39	7.78	9.49	11.14	13.28	14.86	18.47
5	6.63	9.24	11.07	12.83	15.09	16.75	20.52
6	7.84	10.64	12.59	14.45	16.81	18.55	22.46
7	9.04	12.02	14.07	16.01	18.48	20.28	24.32
8	10.22	13.36	15.51	17.53	20.09	21.96	26.12
9	11.39	14.68	16.92	19.02	21.67	23.59	27.88
10	12.55	15.99	18.31	20.48	23.21	25.19	29.59
11	13.70	17.28	19.68	21.92	24.72	26.76	31.26
12	14.85	18.55	21.03	23.34	26.22	28.30	32.91
13	15.98	19.81	22.36	24.74	27.69	29.82	34.53
14	17.12	21.06	23.68	26.12	29.14	31.32	36.12
15	18.25	22.31	25.00	27.49	30.58	32.80	37.70
16	19.37	23.54	26.30	28.85	32.00	34.27	39.25
17	20.49	24.77	27.59	30.19	33.41	35.72	40.79
18	21.60	25.99	28.87	31.53	34.81	37.16	42.31
19	22.72	27.20	30.14	32.85	36.19	38.58	43.82
20	23.83	28.41	31.41	34.17	37.57	40.00	45.32
25	29.34	34.38	37.65	40.65	44.31	46.93	52.62
30	34.80	40.26	43.77	46.98	50.89	53.67	59.70
40	45.62	51.80	55.76	59.34	63.69	66.77	73.40
50	56.33	63.17	67.50	71.42	76.15	79.49	86.66
60	66.98	74.40	79.08	83.30	88.38	91.95	99.61
70	77.58	85.53	90.53	95.02	100.4	104.2	112.3
80	88.13	96.58	101.8	106.6	112.3	116.3	124.8
90	98.65	107.6	113.1	118.1	124.1	128.3	137.2
100	109.1	118.5	124.3	129.6	135.8	140.2	149.5

Classified as Internal | Intern

Chi-squared test of independence

Don't forget…

5. **Conclusion**

Compare p-value to α, relate conclusions back to the context

"We reject the null-hypothesis, and conclude that income and happiness are not independent"

Comments on chi-squared test

- Works well for large random samples
  - Assumption: all $f_e \geq 5$
- For smaller samples: Fisher's exact test
  - Calculations outside scope of this course

- Chi-squared test treats variables as **nominal**
- For ordinal variables, more powerful tests exist (see Chapter 15)

- Web app: https://istats.shinyapps.io/ChiSquaredTest/

Classified as Internal | Intern

2x2: chi-squared vs z-test

- For 2x2 tables, chi-squared test is equivalent to z-test for two groups
- But: chi-squared can only test two-sided H_a

- Example: college alcohol study by Harvard: "Have you engaged in unplanned sexual activities because of drinking alcohol?"
- Change in proportion from 1993 to 2001?

	Yes	No	n
1993	2440 (19.2%)	10268 (80.8%)	12708
2001	1871 (21.3%)	6912 (78.7%)	8783

2x2: chi-squared vs z-test

	Yes	No	n
1993	2440	10268	12708
2001	1871	6912	8783

Chi-squared: first calculate f_e, for which we need the column totals

$$f_e = (row\ total) * (column\ total)/n$$

	Yes	No	Total
1993	2440 (2549)	10268 (10159)	12708
2001	1871 (1762)	6912 (7021)	8783
Total	4311	17180	21490

2x2: chi-squared vs z-test

	Yes	No	Total
1993	2440 (2549)	10268 (10159)	12708
2001	1871 (1762)	6912 (7021)	8783
Total	4311	17180	21490

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e} = 14.3$$

$$df = 1$$

$$p = 0.00016 \text{ (using online calculator)}$$

2x2: chi-squared vs z-test

	Yes	No	Total
1993	2440	10268	12708
2001	1871	6912	8783
Total	4311	17180	21490

z-test: $\hat{\pi}_1 = \frac{2440}{12708} = 0.19$ vs $\hat{\pi}_2 = \frac{1871}{8783} = 0.21$, pooled $\hat{\pi} = \frac{4311}{21490} = 0.2$

$$se_0 = \sqrt{\frac{0.2 * 0.8}{12708} + \frac{0.2 * 0.8}{8783}} = 0.006$$

$z = \frac{(\hat{\pi}_2 - \hat{\pi}_1) - 0}{se_0} = \frac{0.02}{0.006} = 3.78$ (without in-between rounding)

p = 0.00016 (two-sided)

2x2: chi-squared vs z-test

	Yes	No	Total
1993	2440	10268	12708
2001	1871	6912	8783
Total	4311	17180	21490

$$\chi^2 = 14.3$$
$$z = 3.78$$

Squaring the z-value gives the χ^2!

Therefore: two-sided z-test for comparing two proportions is equivalent to a chi-squared test on a 2x2 contingency table

Testing goodness of fit

- Can also use chi-squared test for a single categorical variable when we have 'prior expectations' for its probability distribution

- For example: if birthdates were distributed uniformly at random, we would expect each quarter of the year to have 25% of all births
  - We can test if observations match this expectation

https://istats.shinyapps.io/ChiSquaredTest/

Patterns of association

Universiteit Leiden
The Netherlands

Residuals

- Large χ^2 value implies strong **evidence** of an association, but does not say anything about the **nature** or **strength** of the association
- Investigate by studying the residual in each cell!

- **Residual** $= f_o - f_e$, difference between observed and expected count, positive when there are more observations than expected under H_0
- **Standardized residual** $z = (f_o - f_e)/se$, where se is the standard error of $(f_o - f_e)$, which is different for each cell!!

Computing *se* of $(f_o - f_e)$

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	272 (190)	294 (345)	49 (81)	615
	Average	454 (438)	835 (796)	131 (187)	1420
	Below avg	185 (284)	527 (516)	208 (121)	920
	Total	911	1656	388	2955

$$se = \sqrt{f_e(1 - row\ proportion)(1 - column\ proportion)}$$

First cell: $se = \sqrt{190 * \left(1 - \frac{615}{2955}\right) * (1 - \frac{911}{2955})} = 10.2$

Computing the residual

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	272 (190)	294 (345)	49 (81)	615
	Average	454 (438)	835 (796)	131 (187)	1420
	Below avg	185 (284)	527 (516)	208 (121)	920
	Total	911	1656	388	2955

- First cell: $se = 10.2$
- Residual $= f_o - f_e = 272 - 190 = 82$
- Standardized residual: $z = \frac{f_o - f_e}{se} = \frac{82}{10.2} = 8.1$

The observed number of people that are very happy and have an above average income is 8.1 *se* larger than expected if happiness and income are independent

All standardized residuals

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	272 (8.1)	294 (-4.6)	49 (-4.3)	615
	Average	454 (1.3)	835 (2.9)	131 (-6.0)	1420
	Below avg	185 (-8.5)	527 (0.9)	208 (10.3)	920
	Total	911	1656	388	2955

- Standardized residual is noteworthy if absolute value > 3
- Shows direction of association

- To show pattern of association: calculate all standardized residuals!

Measures of association

Universiteit Leiden
The Netherlands

Measures of association

- Chi-squared test answers: **“is there** an association?”
- Standardized residuals answer: **“how do data differ** from independence?”

- Last question: **“how strong** is the association?”

- Two measures:

1. Difference of proportions
2. Odds ratio -> *self-study* (11.3)

Difference of proportion

- It's in the name: difference of proportion of a certain outcome for two groups
- Example: proportion of people that are very happy, for below vs above average income is $0.44 - 0.20 = 0.24$
- Interpretation: proportion of very happy people among above average income group is 24 percentage points higher than among below average income group

		Happiness			
		Very	Pretty	Not too	*Total*
Income	Above avg	272 (44%)	294 (48%)	49 (8%)	615
	Average	454 (32%)	835 (59%)	131 (9%)	1420
	Below avg	185 (20%)	527 (57%)	208 (23%)	920
	Total	911	1656	388	2955

Example difference of proportion

- Which association is stronger? Political party and opinion, or sex and opinion?

Political party	Approve	Disapprove
Democrats	3%	97%
Republicans	64%	36%

Sex	Approve	Disapprove
Female	24%	76%
Male	27%	73%

- Calculate differences of proportion:
  - For PP: $0.03 - 0.64 = -0.61$
  - For sex: $0.24 - 0.27 = -0.03$

- Association with political party is stronger!

Relative risk

- Rather than the difference, the ratio of proportions can also be useful (also called the "relative risk")

- *Example*: Comparing proportions who report being very happy, for those of above average income to those of below average income:

$$0.44/0.20 = 2.2$$

- *Interpretation*: proportion of very happy people is 2.2x as high for people with above average income than for people with below average income

Difference vs ratio

- Risk of lung cancer
  - Smokers: 3%
  - Non-smokers: 0.2%

- Risk of lung cancer is 2.8 percentage points higher for smokers than for non-smokers (difference of proportions)
- Risk of lung cancer is 15x higher for smokers than for non-smokers (ratio of proportions, relative risk)

Evidence vs strength of association

- The chi-squared test merely analyzes the extent of evidence that there is an association
  - Does not tell us the ***nature*** of the association
  - Does not tell us the ***strength*** of association

- A large chi-squared test statistic and small *P*-value indicates *strong evidence* of association but not necessarily a *strong association*

Recall: *statistical significance* not the same as *practical significance!*

Effect of n on statistical significance

n = 50

	yes	no
Group 1	15	10
Group 2	10	15

$\chi^2 = 2$
$p = 0.18$

n = 100

	yes	no
Group 1	30	20
Group 2	20	30

$\chi^2 = 4$
$p = 0.048$

n = 500

	yes	no
Group 1	150	100
Group 2	100	150

$\chi^2 = 20$
$p < 0.001$

All tables have the same proportions (and thus difference of proportions):

$$\hat{\pi}_1 = 15/(15 + 10) = 0.6$$
$$\hat{\pi}_2 = 10/(10 + 15) = 0.4$$
$$\hat{\pi}_1 - \hat{\pi}_2 = 0.6 - 0.4 = 0.2$$

A weak association can still give small p-values for large n!

Example: small p-value, weak association

- $\chi^2 = 8$
- $p = 0.005$

	yes	no
Group 1	5100	4900
Group 2	4900	5100

- $\hat{\pi}_1 - \hat{\pi}_2 = 0.51 - 0.49 - 0.02$

- Very strong **evidence** of association, but the association itself is quite **weak**

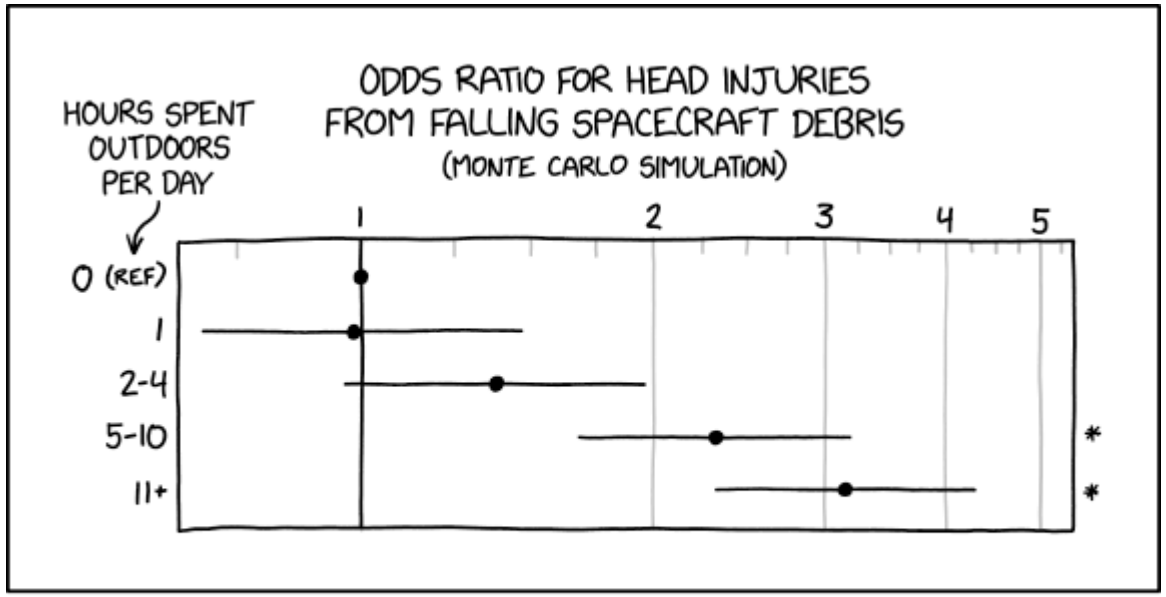
Wrapping up

Universiteit Leiden
The Netherlands

Wrapping up

- Read Chapter 11 of the book

- Today:
  - Two categorical variables
- Next week: two quantitative variables! *Regression and correlation*



OUR NEW STUDY SUGGESTS THAT SPENDING MORE THAN 5 HOURS OUTSIDE SIGNIFICANTLY INCREASES YOUR RISK OF HEAD INJURY FROM SPACECRAFT DEBRIS, SO TRY TO LIMIT OUTDOOR ACTIVITIES TO 4 HOURS OR LESS.

https://xkcd.com/2599/